



Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

OBSIDYN-OS

Himesh Saini
2210990410

Krithik Saini
2210990522

Hirdesh Mittal
2210990522

Supervised By
Dr. Gurpreet Singh
Associate Professor

Department of Computer Science and Engineering,
Chitkara University, Punjab

A large amount of personal and professional information is stored on individual computer systems today. These systems are widely used for everyday tasks, but they are not designed to keep sensitive data strictly isolated.

- In common environments, files can be accessed, copied, or recovered with basic methods, which makes data handling less controlled.
- At the same time, the level of technical knowledge required to properly manage data security is often higher than what most users have.
- There is also a need for an approach where access to data depends on the identity and control of the user, rather than only on basic system-level permissions
- Because of this, there is a growing need for a more controlled way of handling sensitive data within a system.

In most systems, security is tied to a single point — access. Once that boundary is crossed, control over data is effectively lost. A compromised password, shared system, or unintended access directly exposes all stored files without any internal restriction.

- If authentication is bypassed or credentials are leaked, the entire dataset becomes readable
- Data remains usable outside the system — copying a file is enough to break control
- Extracted or protected files, if not deeply secured, can still be inspected using low-level tools like hex editors
- File deletion is unreliable; data can be recovered with standard recovery tools
- The system assumes “access = trust”, with no mechanism to enforce control after entry

The scope of this system is limited to a local environment where sensitive data is handled independently of default operating system behavior.

The objective is to establish a controlled method of data handling where access alone does not define usability.

- Introduce a controlled execution environment for handling sensitive data
- Reduce reliance on single-point authentication mechanisms
- Ensure data does not remain directly interpretable at storage level
- Maintain control over how data is accessed and processed
- Limit the impact of unauthorized system access on stored information
- Enforce structured handling of data within a defined environment

6. Methodology

Design Approach

The system is developed using a security-first approach where all design decisions are based on potential threat scenarios. Instead of focusing only on functionality, the system assumes that both the operating environment and user credentials can be compromised.

Threat Modeling

Two main threat categories are considered. The first includes remote threats such as malware, data exfiltration, and memory scraping. The second includes local threats such as physical access, unauthorized file inspection, and recovery attempts. The system is designed to handle both cases by minimizing data exposure and restricting direct access.



Architectural Strategy

A decoupled architecture is implemented where the user interface and execution engine operate as separate processes. Communication between these components is handled through a controlled IPC mechanism using structured data exchange. This separation ensures that vulnerabilities in one layer do not directly affect others.

Security Principles

The system avoids reliance on the host operating system for security. Data is never stored in plaintext form, and all operations are validated before execution. Sensitive information such as keys is maintained only in memory and is removed after use.

Architecture Overview

The system follows a layered architecture that separates user interaction, communication, processing, and storage. This design ensures that each component performs a specific role without directly exposing sensitive data.

Architecture Layers

Presentation Layer

Handles all user interactions and input processing. It does not perform any sensitive operations or directly access stored data.

Communication Layer

Transfers structured requests between the interface and backend using a controlled communication channel.

Execution Layer

Acts as the central controller that validates incoming requests and dispatches them to appropriate modules.
storage.



Authentication Layer

Verifies user identity through multiple validation mechanisms before allowing access.

Cryptographic Layer

Handles encryption and decryption using standardized algorithms to ensure data confidentiality and integrity.

Steganography Layer

Enables data to be embedded within media files, allowing hidden storage and controlled extraction.

Defense Layer

Monitors system activity, detects anomalies, and provides mechanisms to respond to unauthorized actions.

OS Interaction Layer

Manages interaction with the file system while maintaining hidden and controlled data

Instead of relying on the operating system to manage sensitive data, the system introduces a controlled layer where all data interactions pass through a dedicated engine. Data is not treated as regular files and is only interpreted within this environment.

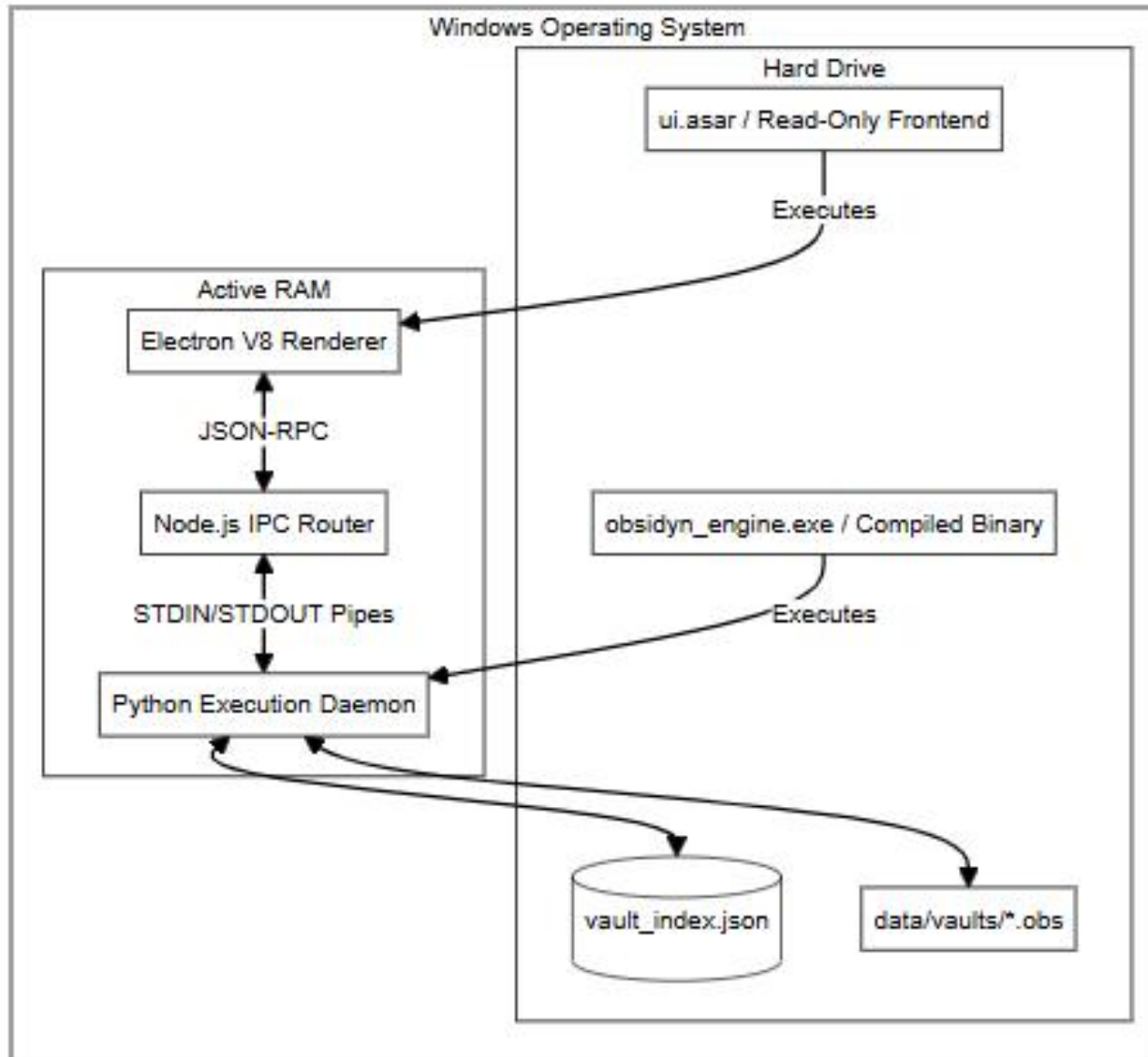
- A central engine handles all operations related to authentication, storage, and data access
- The user interface acts only as an interaction layer, without direct access to underlying data
- Data is transformed into a non-readable form before storage and is only interpreted within the system
- All access requests are validated and routed through a controlled execution flow
- Direct interaction with raw data at system level is avoided

System Architecture

The system is organized into a layered architecture to separate interaction, communication, execution, and data handling.

- Presentation Layer
 - Provides the user interface for interaction and input handling without direct access to sensitive data
- Communication Layer
 - Transfers structured requests between the interface and the execution engine using a controlled channel
- Core Execution Engine
 - Acts as the central controller responsible for validating and dispatching all operations
- Subsystem Layer
 - Contains dedicated components for authentication, data processing, and system control
- Storage Layer
 - Maintains data in a controlled and non-interpretable format
- Runtime Layer
 - Handles session-based operations and temporary execution states in memory

This layered design ensures that all data interactions are processed through controlled pathways, maintaining separation between system components and reducing direct exposure.



This system introduces a shift from traditional access-based protection to controlled data interaction, where all operations are governed through a centralized execution model. Instead of relying on the operating system to enforce security, the system establishes its own controlled handling mechanism for sensitive data.

1. Centralized Execution Control

All system operations are routed through a single execution engine that validates and dispatches requests.

- Prevents direct interaction with underlying data
- Ensures consistent enforcement of system rules
- Eliminates uncontrolled file-level operations

This creates a single point of control for all data interactions



2. Session-Oriented Security Model

Access to data is not permanent and is bound to an active session.

A session is established only after successful identity verification

Sensitive operations are allowed only within session context

Session state is maintained in volatile memory

This limits data access to time-bound, controlled execution states

3. Context-Based Data Interpretation

Stored data does not remain directly meaningful outside the system.

Data is transformed before storage

Interpretation depends on system-controlled processes

External tools cannot directly derive usable content

Data is context-dependent, not self-contained



4. Layered System Isolation

The system enforces separation between interaction, execution, and storage layers.

UI handles only input and output

Core engine processes all logic

Storage layer remains isolated from direct access

This reduces exposure and enforces controlled pathways

5. Controlled Data Lifecycle Management

All stages of data handling are governed by system logic.

Data creation, storage, access, and removal follow defined flows

Default OS behavior is not relied upon for sensitive operations

Data is never exposed in raw form during processing

Ensures consistent and predictable handling of sensitive data

Authentication Subsystem

The authentication process is designed to validate identity through multiple stages. It combines traditional password verification with behavioral analysis such as typing patterns. Additional recovery mechanisms are available to ensure access in case of failure, while maintaining security.

Cryptographic Core

The system uses AES-256-GCM encryption to ensure both confidentiality and data integrity. Encryption and decryption processes are performed in memory, preventing exposure of plaintext data on the disk. Key derivation techniques are used to convert user credentials into secure cryptographic keys.

Steganography Module

This module allows data to be embedded within image files using bit-level manipulation. The process ensures that embedded data does not affect the visible structure of the image. A sanitization mechanism is also provided to remove hidden data completely when required.

Secure Vault System

Data is stored in an encrypted format using a structured storage mechanism. Each file is individually encrypted and managed, preventing large-scale data compromise. Metadata is maintained separately to allow efficient retrieval without exposing actual content.

Active Defense Mechanism

The system includes proactive defense features such as generating decoy files to mislead attackers. These decoys resemble real data and increase the difficulty of identifying actual sensitive information.

Monitoring System

A continuous monitoring process tracks system-level activities related to stored data. It detects unauthorized modifications, deletions, or additions and provides alerts for further action.

Secure Deletion

Instead of relying on standard deletion, the system overwrites data multiple times to ensure it cannot be recovered. This prevents forensic tools from retrieving previously stored information.



IPC and Session Management

All system operations are routed through a central engine using structured communication. Session keys are stored only in memory and are removed once the session ends, ensuring no per

The system was evaluated through practical testing scenarios. Stored data was inspected using low-level tools and appeared as non-readable content. Attempts to recover deleted files using standard recovery software were unsuccessful. Authentication tests confirmed that incorrect behavioral inputs were rejected even when passwords were correct. Steganography operations successfully hid and removed data without leaving detectable traces.



12:38:18

OBSIDYN

SILENT VAULT. BEHAVIORAL GATE. MINIMAL TRACE.

MASTER KEY

Enter your master key

SECURITY PROFILE

PERSONAL

WORK

PUBLIC

ESTABLISH SESSION

☒ PVS-PASS BYPASS

No master identity enrolled yet. First session will establish Rhythm Lock.

Privacy-first runtime active



Visual Recovery Override

CAPTURE LIVE FACE AND GESTURE, THEN SUBMIT THEM FOR VERIFICATION.



1 Capture face

FRONT-FACING FRAME IN CLEAR LIGHT

2 Capture gesture

HOLD YOUR CHOSEN HAND SIGNATURE
STEADY

3 Enable recovery

SAVE BOTH CAPTURES INTO SECURE
RECOVERY ENROLLMENT



CAPTURE FACE

CAPTURE GESTURE

FACE FRAME



GESTURE FRAME



RESET FRAMES

WAITING FOR FACE AND GESTURE CAPTURE.

SUBMIT & VERIFY

OBSIDYN

SYS_TIME
12:42:34

AUTO_LOCK
09:03

PROFILE
PERSONAL

✓ APPLY





BRDashboard

OPOperator

VTVault Manager

DCDecoy Ops

SGSignals

XRSecure Shred

PXSteganography

PVPVS Pass

PMPC Manager

CTSettings

OPERATIONAL BRIEFING

Session posture, seal counts, and current defensive state.

Encryption

AES-256-GCM

ACTIVE PROTOCOL

Protected Assets

0

FILES ENCRYPTED

Security Level

GUARDED

CURRENT STATUS

Auto-Lock

09:03

TIME REMAINING

System Vitals

12

16 GB

5.4 GB

51h 38m



OBSIDYN

SYS_TIME
12:44:09

AUTO_LOCK
07:29

PROFILE
PERSONAL

✓ APPLY

≡

BR Dashboard

OP Operator

VT Vault Manager

DC Decoy Ops

SG Signals

XR Secure Shred

PX Steganography

PV PVS Pass

PM PC Manager

CT Settings

OPERATOR CONTROL

Private dossier, note storage, recovery enrollment, and behavioral controls.

Operator Dossier

Single-user identity record with controlled disclosure, protected image identity, and sealed profile controls.

CANCEL

SEAL DOSSIER

No dossier sealed yet. Fill the fields once, seal the record, then manage it through view and update controls.

OP

Notes Vault

Headings stay visible as a horizontal rail. Full content stays inside the sealed note workspace.

UNCONFIGURED



OBSIDYN

SYS_TIME

12:45:35

AUTO_LOCK

06:03

PROFILE

PERSONAL

✓ APPLY

BR

Dashboard

OP

Operator

VT

Vault Manager

DC

Decoy Ops

SG

Signals

XR

Secure Shred

PX

Steganography

PV

PVS Pass

PM

PC Manager

CT

Settings

Operations Protocol

Encryption:

AES-256-GCM. Original references are securely shredded post-sealing to prevent forensic recovery.

Extraction:

Restores the payload precisely without leaving temporary artifacts on the host.

[+] SEAL FILE

[+] SEAL FOLDER

[SYNC] REFRESH

Encrypted Assets

0 ITEMS

Vault inventory is clear.

Seal a file or folder to create a hidden record.



OBSIDYN

SYS_TIME

12:46:47

AUTO_LOCK

04:51

PROFILE

PERSONAL

✓ APPLY





BR Dashboard

OP Operator

VT Vault Manager

DC Decoy Ops

SG Signals

XR Secure Shred

PX Steganography

PV PVS Pass

PM PC Manager

CT Settings

DECEPTION OPERATIONS

Deploy bait vaults and watch honeyfiles for interference.

SELECT DEPLOYMENT DIRECTORY

SEED DECOY VAULT

REFRESH STATUS

DELETE ALL DECOY FILES

CLEAR HISTORY

DOWNLOAD MEMORY LOG

Decoy Vaults

1

ACTIVE DEPLOYMENTS

Honey Alerts

4

RECORDED TRIGGERS

Email Channel

STANDBY

FILE_OPENED AT 12:46:44

Memory Log

4

Target Root

D:\DEMOoooooooooooooooooooo



OBSIDYN

SYS.TIME
12:47:50

AUTO_LOCK
03:48

PROFILE
PERSONAL

✓ APPLY

≡

🔒

BR Dashboard

OP Operator

VT Vault Manager

DC Decoy Ops

SG Signals

XR Secure Shred

PX Steganography

PV PVS Pass

PM PC Manager

CT Settings

PIXEL VEIL

Embed or extract concealed payloads with encrypted transport.

🔑 Hide Data

🔓 Extract Data

🔍 Scan Image

🗑️ Sanitize

DATA FILE TO HIDE

No file selected

Browse

CARRIER IMAGE

No image selected

Browse

🔑 PASSWORD (OPTIONAL)

Leave empty for no pas

💾 SAVE AS

Auto-generated if empty

Choose...

📁 CAPACITY CHECK

Select files to check capacity

🔑 ENCODE & HIDE

🔑 How Pixel Veil Works

- **Compression:** Files are ZIP-compressed before hiding (40-60% size reduction)
- **Encryption:** AES-256-GCM encryption if password is set
- **Steganography:** Data hidden in LSB of RGB pixels using adaptive algorithm
- **Undetectable:** Output image is visually identical; PSNR > 40dB
- **OBSIDYN Signature:** Only our system can detect and extract hidden data



PIXEL VEIL

Embed or extract concealed payloads with encrypted transport.

Hide Data

Extract Data

Scan Image

Sanitize

IMAGE TO SANITIZE

No image selected

Browse

SAVE AS

Auto-generated if empty

Choose...

STRIP HIDDEN DATA

How Pixel Veil Works

- **Compression:** Files are ZIP-compressed before hiding (40-60% size reduction)
- **Encryption:** AES-256-GCM encryption if password is set
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Capacity Guide

640×480
~900 KB

1920×1080
~5.5 MB

3840×2160 (4K)
~22 MB

8MP Photo
~16 MB



 **OBSIDYN**

SYS_TIME
12:51:54

AUTO_LOCK
09:54

PROFILE
PERSONAL

✓ APPLY

⌵

🔒

BR Dashboard

OP Operator

VT Vault Manager

DC Decoy Ops

SG Signals

XR Secure Shred

PX Steganography

PV PVS Pass

PM PC Manager

CT Settings

CONTROL

Behavioral lock and session configuration.

Session Runtime

MINUTES OF INACTIVITY

10

DEFAULT PROFILE

PERSONAL

WORK

PUBLIC

Runtime settings loaded and synchronized.

Visual Recovery Access

ALLOW VISUAL RECOVERY ON LOGIN
SHOW FACE + GESTURE RECOVERY ON THE LOGIN SCREEN WHEN AN ENROLLMENT EXISTS.
☒

No visual recovery signature stored.

OPEN OPERATOR RECOVERY

Layout Profiles

SAVED LAYOUTS

Custom Auto (12:43 PM)

LOAD

SAVE CURRENT LAYOUT AS...

Privacy Surface

PRIVACY MODE
PREFER QUIETER RUNTIME BEHAVIOR AND LOWER VISIBLE TRACE.
☐

STORE FULL PATHS
KEEP ORIGINAL SOURCE PATHS IN METADATA INSTEAD OF MASKING THEM.
☐

DECOY EMAIL ALERTING
SEND A SILENT EMAIL ALERT WHEN A DECOY VAULT IS ACCESSED.
☐

Interface Aesthetics

LOGIN BINARY FIELD
SHOW THE ANIMATED BINARY ATMOSPHERE ON THE LOGIN SCREEN.
☐

REDUCED MOTION
TRIM INTERFACE ANIMATION AND MOTION INTENSITY FOR A QUIETER SURFACE.
☐

COMPACT INTERFACE MODE
CONDENSE PADDING AND MARGINS TO MAXIMIZE DATA DENSITY ON SCREEN.
☐



BR Dashboard

OP Operator

VT Vault Manager

DC Decoy Ops

SG Signals

XR Secure Shred

PX Steganography

PV PVS Pass

PM PC Manager

CT Settings

OBLIVION SHREDDER

Controlled destruction for assets you choose to erase.



IRREVERSIBLE DESTRUCTION

Files shredded with this tool cannot be recovered by any means. This includes forensic recovery tools, data recovery software, and law enforcement techniques.



Drop Files to Shred

or



Select File to Destroy

Supports: Documents, Images, Videos, Archives, Executables

Destruction Method



DOD 5220.22-M

3-PASS OVERWRITE (RECOMMENDED)

The system was tested under multiple scenarios, including simulated attacks and unauthorized access attempts. Black-box testing was used to evaluate system behavior without internal knowledge. The system maintained stability and consistently enforced security constraints under different conditions.

- Black-Box Security Testing
- Hex-Level Data Inspection
- Ciphertext Tampering Test
- Data Recovery Testing
- Memory-Based Attack Testing
- Authentication Bypass Testing
- Steganography Validation Testing
- Steganographic Sanitization Testing
- Decoy (Honey File) Testing
- Monitoring and Intrusion Detection Testing
- IPC Security Testing
- System Stability Testing

Conclusion

The system demonstrates that data security can be improved by controlling how data is handled, not just by restricting access. By combining multiple mechanisms into a unified environment, it ensures that sensitive data remains protected and controlled under various conditions.

Future Scope

Future improvements may include advanced biometric authentication, enhanced data hiding techniques, support for distributed environments, and optimization of system performance for larger datasets.

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Thank You